

Zainab Tairu

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Bio

Undergraduate researcher with expertise in NLP, applying it to molecular structures and health applications in low-resource settings

Education

Bachelor of Science in Chemical Engineering Jan 2022–2027
Obafemi Awolowo University, Osun, Nigeria
Relevant Courses: Thermodynamics (A), Python (A), Mathematical Methods/Linear Algebra (B)

Work Experience

CoMMA Lab, University of Texas, USA Jul 2025–Present
Research Intern

- Wrote a survey paper on Machine learning in Polymer science

NNPC Investment Management Services, Lagos, Nigeria Oct 2025–Dec 2025
Process Intern

- Conducted exploratory research integrating well log and seismic data for reservoir characterization.
- Designed an ETL pipeline automating daily production reports from field operators, improving data accessibility and reporting efficiency.
- Explored ML applications for predictive maintenance of factory equipment.

The Innoverse, South Africa Aug 2024–Jul 2025
NLP Intern

- Trained an intelligent chatbot leveraging Retrieval-Augmented Generation (RAG) with Neo4j Knowledge Graphs for contextual retrieval.
- Fine-tuned low-resource text-to-speech (TTS) models for English and Zulu to enhance multilingual accessibility.
- Explored agent-based AI systems for dynamic knowledge-driven conversations.

Publications

Papers and Preprints

- Machine Learning Models and Their Complexities in Polymer Informatics: A Systematic Review (**Expected submission Feb 2026**).
- Evaluating Retrieval-Augmented Generation for Medication Question Answering on Nigerian Drug Labels in Yorùbá (**Submitted to a workshop at EACL**).

Poster Presentations

- Hot and Sustainable: AI-Guided Design of Thermostable Lipases, Deep Learning Indaba 2025
- Dual AI Architectures for Accessible Medication Management: OCR-LLM vs. VLM, Deep Learning Indaba 2025
- Reservoir Characterization with Machine Learning, SPE OAU Hackathon 2025

Projects

MediVoice May 2025

- Built a voice-based pipeline enabling visually impaired users to manage medication schedules independently.

UlcerMate Jun 2025

- Developed an AI-powered pipeline to help stomach ulcer patients track symptoms and medications while detecting personal ulcer triggers using machine learning.

Malaria Blood Smear Classifier

Jun 2025

- Built a CNN algorithm achieving 99% accuracy for malaria diagnosis from blood smear images.

RFE Risk Analyzer

Jul 2025

- Built a USCIS-style risk analyzer using DeepSeek LLM and Ollama to simulate adjudicator feedback for EB-1A petitions.

Conferences, Workshops, Other Research Activities

Deep Learning Indaba, Kigali, Rwanda (poster presented)	2025
SPE Hackathon, Nigeria (team lead)	2025
Deep Learning Indaba, Dakar, Senegal (attended)	2024
DSN AI Bootcamp, Nigeria (participant)	2024
DS4Health 2025	

Teaching Experience

DSN OAU	May 2025–Jun 2025
Deep Learning Tutor	
• Taught foundational deep learning concepts, including building Convolutional Neural Networks (CNNs) using several libraries.	
Star Girls Africa	Jun 2024–Aug 2024
Data Science Tutor	
• Instructed over 100 students in Python programming and supervised their first end-to-end machine learning model training.	

Relevant Extracurricular Experience

Lead, DSN OAU	Dec 2025 – Present
• Driving the growth and direction of the DSN OAU community by building inclusive structures	
• Introduced a research track to support early-stage researchers.	
Deep Learning Lead, DSN OAU	Jan 2025– Nov 2025
• Coordinated talks and invited speakers on advanced and ethical deep learning topics.	
Public Relations Officer, Space Club OAU	Jan 2025–Oct 2025
• Managed outreach campaigns and collaborations to promote space and AI education on campus.	
Member, SPE OAU	Jan 2022–Present
• Participated in hackathons and applied machine learning to practical problems in the petroleum engineering space.	

Skills

- Advanced • Proficient • Moderate
- **Programming:** Python •, SQL •, Cypher •, MATLAB •, Bash •
 - **Machine Learning & AI:** Deep Learning •, Transformers •, RAG •, LLMs •, CNNs •
 - **Data Analysis & Visualization:** NumPy •, Pandas •, JAX •, Matplotlib •, Seaborn •, Plotly •
 - **Natural Language Processing:** NER •, Sentiment Analysis •, Prompt Engineering •, Rasa •, LangChain •
 - **Scientific Computing & Bioinformatics:** Molecular Modeling •, Protein Sequence Analysis •, Computational Chemistry •
 - **Frameworks & Libraries:** PyTorch •, TensorFlow •, Hugging Face •, scikit-learn •
 - **Databases:** Neo4j •, SQL •
 - **Cloud, Deployment & DevOps:** AWS •, Docker •, FastAPI •, Streamlit •, Gradio •, GitHub Actions •
 - **Version Control & Collaboration:** Git •, GitHub •